AVIKAM JHA

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SUMMARY

AI/ML researcher and engineer (MS Computer Engineering, ASU, 4.0 GPA) working on human-centered robot learning and LLM interpretability. Thesis research spans human-subjects evaluation of robot demonstration data collection and sparse-autoencoder interpretability of LLM planning behavior. Brings 2+ years of production infrastructure engineering (Kubernetes, GCP/AWS) and hands-on CUDA/PyTorch systems work.

EDUCATION

Arizona State University — M.S. Computer Engineering (Thesis), GPA 4.0 Aug 2024 – Dec 2026

- Thesis committee: Prof. Lindsay Sanneman, Prof. Ransalu Senanayake, Prof. Nakul Gopalan; defense Nov 2026
- Coursework: Artificial Intelligence, Foundations of Algorithms, Probability and Random Processes, Data Processing at Scale, Computer Architecture, VLSI Design, Digital Verification and Test

National Institute of Technology, Delhi — B.E. Electronics and Communication Engineering Aug 2022

- Coursework: Pattern Recognition and Machine Learning, Neural Networks, Data Structures and Algorithms, Digital Signal Processing, Control Theory

THESIS RESEARCH

Human Workload Across Robot Demonstration Data Collection Modalities ASU · 2026

- Designed an IRB-approved, within-subjects human study comparing five data collection modalities for robot learning: SpaceMouse teleoperation, leader-arm teleoperation, VR teleoperation, head-mounted video and data gloves.
- Measures operator workload, fatigue and ergonomics on two manipulation tasks (cloth folding, cup stacking), run in a single session per participant.
- Builds on a 27-paper systematic review (HRI 2027 submission) that identified the human-subject and data-efficiency experiments the field still lacks.

Interpretability of LLM Planning via Sparse Autoencoders ASU · 2025 – 2026

- Built a pipeline with SAELens and TransformerLens to extract SAE features from an LLM on automated planning problems, surface top-activating prompts and auto-label features with LLM summarization.
- Designed a human-in-the-loop study testing whether showing SAE feature labels helps people predict when the LLM will fail on complex planning tasks.
- Augmented prompts with top-n activating feature information to analyze how predictable LLM planning performance is.

PUBLICATIONS

Robotics Data Collection Modality Survey — A. Jha, R. Sadhwani, L. Sanneman. Sep 2026

Submitted to ACM/IEEE HRI 2027 (Theory and Methods), under review

- Systematic review of 27 post-2022 studies comparing teleoperation, wearable capture and human-video modalities for robot demonstration data.
- Screened 685 records (PRISMA) to a corpus covering 392 participants; verified every extracted figure against sources, which falsified and revised four initial gap claims.
- Built a modality taxonomy and a nine-gap analysis specifying the experiments the field still needs.

AI / ML PROJECTS

Custom CUDA Kernels for PyTorch ASU · 2025

- Wrote CUDA kernels (shared memory, custom launches) for core tensor ops and integrated them into PyTorch, including forward and backward passes for Linear and ReLU.
- Exposed standalone CUDA code to Python through ctypes bindings, removing framework overhead on heterogeneous CPU/GPU systems.

Reinforcement Learning for Automated Book Organization ASU · 2025

- Implemented tabular Q-learning with an epsilon-greedy policy and 1% decay to balance exploration and exploitation; analyzed convergence via cumulative-reward curves.

Automated Planning and Heuristic Search ASU · 2025

- Wrote PDDL domains and problems for robot pick-and-move tasks solved with FF; compared A*, GBFS, UCS and BFS in a Turtlebot environment.
- Designed a custom A* heuristic that outperformed Manhattan distance on complex grids.

ASIC Acceleration of Graph Convolutional Networks ASU · 2025

- Designed and verified SystemVerilog matrix-multiply and convolution modules for GCNN inference; ran DRC/LVS and post-layout simulation on 7nm FinFET.

EXPERIENCE

Park+ (Parviom Technologies) — SDE Intern → SDE 1 → SDE 2, Gurugram, India Mar 2022 – Jul 2024

- Owned Linux and Kubernetes infrastructure for a 60+ microservice platform.
- Led the AWS-to-GCP migration of Elasticsearch, Redis, MongoDB and Kubernetes with under 2 hours of downtime, cutting operating costs 30% and improving system performance 25%.
- Built Shell and Python automation that made incident response 20% faster.

SKILLS

- **ML and interpretability:** PyTorch, NumPy, pandas, SAELens, TransformerLens, sparse autoencoders, concept bottleneck models, OpenAI API, LLM evaluation, reinforcement learning, PDDL planning, heuristic search
- **GPU and systems:** CUDA kernel development, PyTorch backend integration, ctypes bindings, CPU/GPU heterogeneous computing
- **Robotics and HRI:** teleoperation (SpaceMouse, leader arm, VR), wearable and video demonstration capture, human-subjects study design, workload measurement
- **Languages:** Python, C++, C, CUDA, Bash, MATLAB, SystemVerilog
- **Infrastructure:** Kubernetes, Docker, AWS, GCP, Azure, Jenkins, Git, Linux
- **Research methods:** systematic review (PRISMA), research synthesis, IRB protocols, LaTeX